# Engineering Mechanics: ME101

# Lecture 16-17: Statics


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# Contents

# Dimension & Unit of Work

(Force) x (Distance) → Joule (J) = N.m

→ Work done by a force of 1 Newton moving through a distance of 1 m in the direction of the force

→ Dimensions of Work of a Force and Moment of a Force are same though they are entirely different physical quantities.

→ **Work** is a scalar given by dot product; involves product of a force and distance, both measured along the same line

→ **Moment** is a vector given by the cross product; involves product of a force and distance measured at right angles to the force

→ Units of Work: Joule

→ Units of Moment: N.m

# Virtual Work *(contd.)*

## Method of Virtual Work

- Previous methods (FBD, $\sum F$, $\sum M$) are generally employed for a body whose equilibrium position is known or specified
- For problems in which bodies are composed of interconnected members that can move relative to each other.
- → various equilibrium configurations are possible and must be examined.
- → previous methods can still be used but are not the direct and convenient.
- **Method of Virtual Work** is suitable for analysis of multi-link structures (pin-jointed members) which change configuration
- effective when a simple relation can be found among the disp. of the pts of application of various forces involved
- → based on the concept of work done by a force
- → enables us to examine stability of systems in equilibrium


Scissor Lift Platform

# Virtual work


- The **principle of virtual work** is particularly useful when applied to the solution of problems involving the equilibrium of machines or mechanisms consisting of several connected members.

# Introduction

- **Principle of virtual work :** If a particle, rigid body, or system of rigid bodies which is in equilibrium under various forces is given an arbitrary **virtual displacement**, the net work done by the external forces during that displacement is zero.

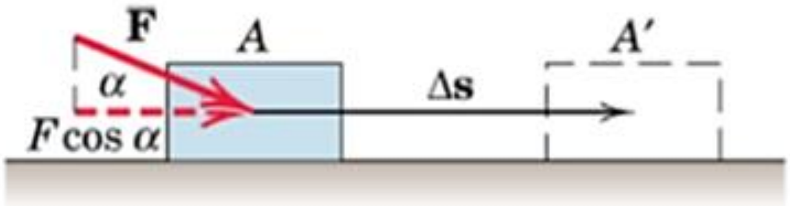
- The principle of virtual work is particularly useful when applied to the solution of problems involving the equilibrium of machines or mechanisms consisting of several connected members.

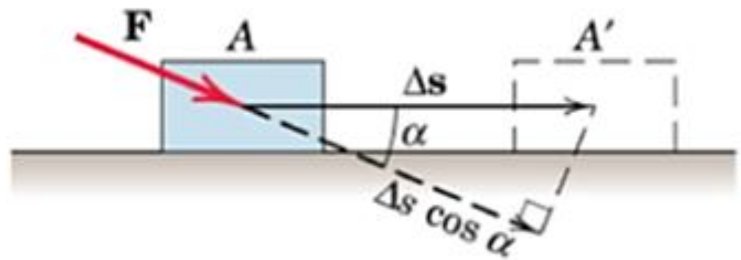
- If a particle, rigid body, or system of rigid bodies is in equilibrium, then the derivative of its potential energy with respect to a variable defining its position is zero.

- The stability of an equilibrium position can be determined from the second derivative of the potential energy with respect to the position variable.

# Virtual Work *(contd.)*

## Work done by a Force (*U*)

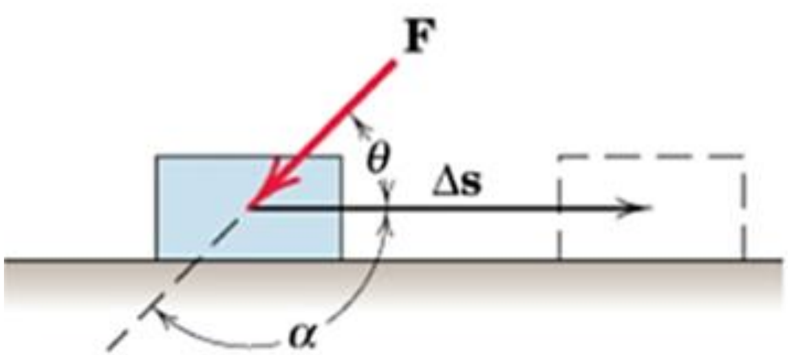



*U* = work done by the component of the force in the direction of the displacement times the displacement

$U = (F\cos\alpha)\,\Delta s$ or $U = F(\Delta s\cos\alpha)$

Since same results are obtained irrespective of the direction in which we resolve the vectors → Work is a scalar quantity




+*U* → Force and Disp in same direction

- *U* → Force and Disp in opposite direction

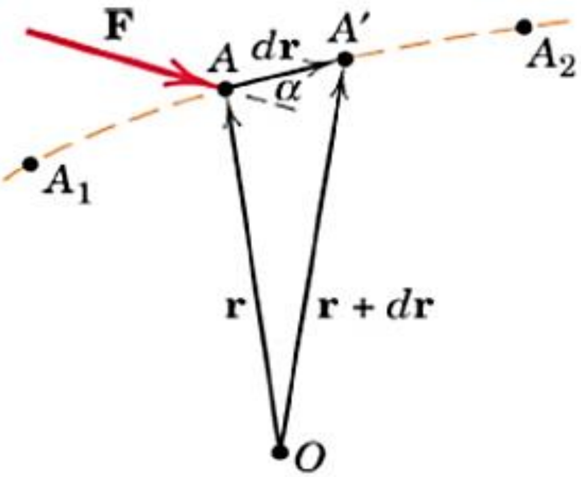
$U = (F\cos\alpha)\,\Delta s = -(F\cos\theta)\,\Delta s$

# Virtual Work *(contd.)*

## Work done by a Force ($U$)

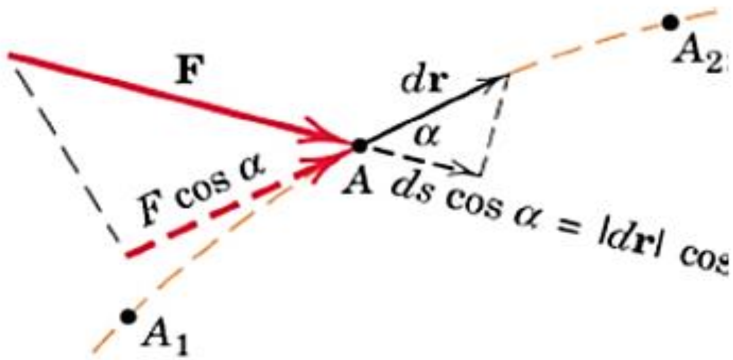
Generalized Definition of Work

Work done by **F** during displacement $d\mathbf{r}$




$$\boxed{dU = \mathbf{F}\cdot d\mathbf{r}}$$

→ $dU = F\, ds \cos\alpha$



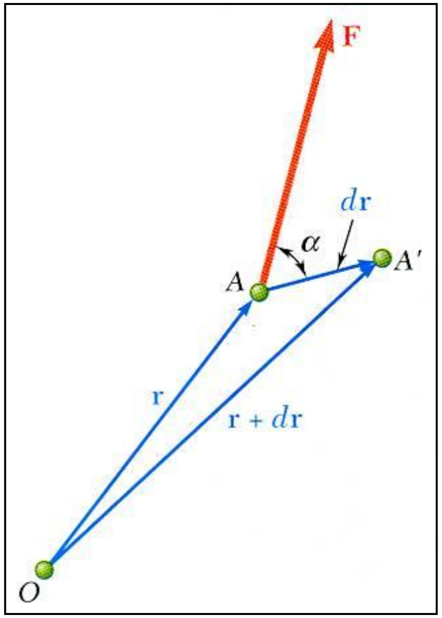

Expressing **F** and $d\mathbf{r}$ in terms of their rectangular components

$$dU = (\mathbf{i}F_x + \mathbf{j}F_y + \mathbf{k}F_z)\cdot(\mathbf{i}\,dx + \mathbf{j}\,dy + \mathbf{k}\,dz)$$
$$= F_x\,dx + F_y\,dy + F_z\,dz$$

Total work done by **F** from $A_1$ to $A_2$

$$U = \int \mathbf{F}\cdot d\mathbf{r} = \int (F_x\,dx + F_y\,dy + F_z\,dz)$$ → $$U = \int F\cos\alpha\, ds$$

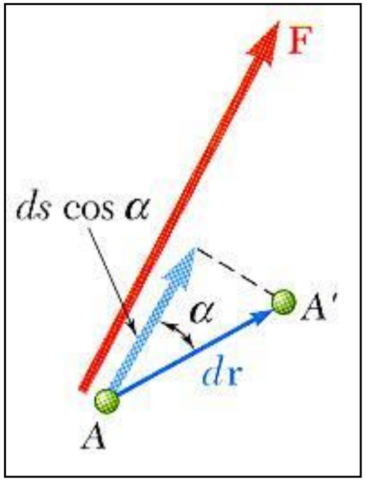
# Work of a Force

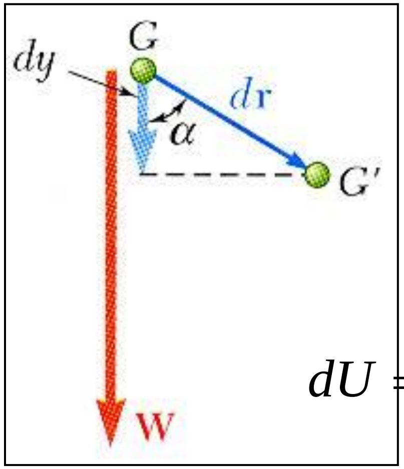



$d U=\vec{F}\cdot d\vec{r}$ = work of the force $\vec{F}$ corresponding to the displacement $d\vec{r}$

$$d U=F\,ds\cos\alpha$$

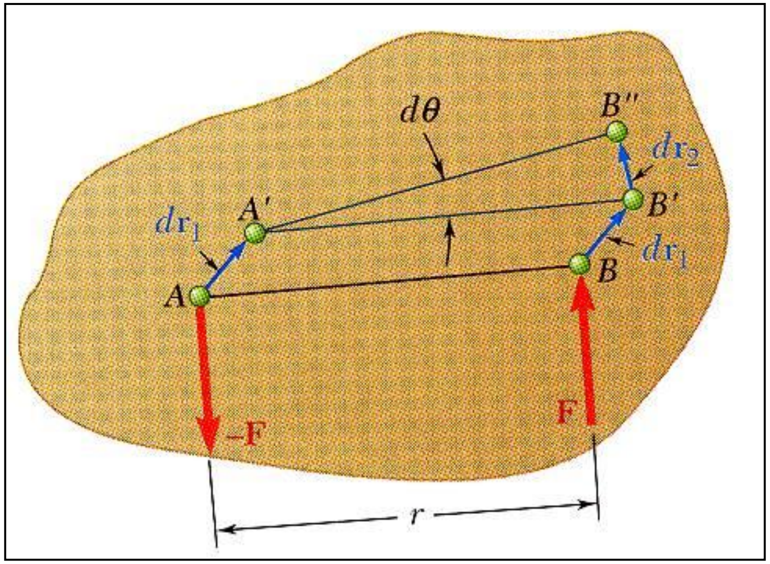
$\alpha=0,\ dU=+F\,ds$ $\alpha=\pi,\ dU=-F\,ds$ $\alpha=\frac{\pi}{2},\ dU=0$




$$dU=Wdy$$

# Work of a Couple



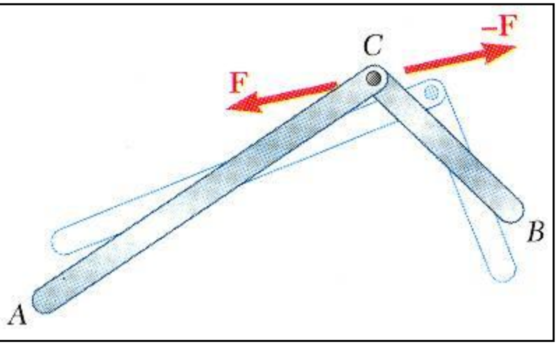

❖Small displacement of a rigid body:

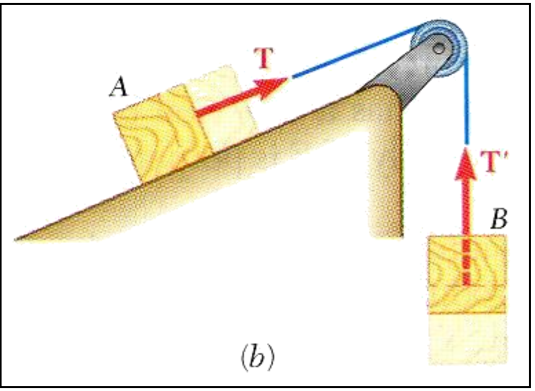
✓translation to *A'B'*

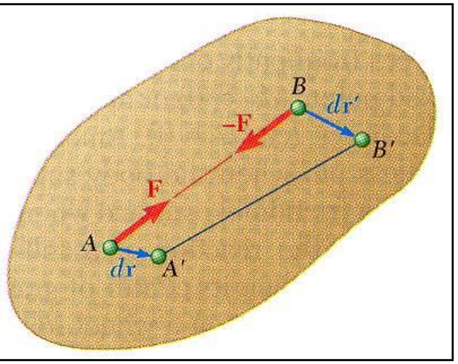
✓rotation of *B'* about *A'* to *B''*

$$\begin{aligned} W &= -\vec{F}\cdot d\vec{r}_1 + \vec{F}\cdot\left(d\vec{r}_1 + d\vec{r}_2\right) \\ &= \vec{F}\cdot d\vec{r}_2 = F\,ds_2 = F\,r d\theta \\ &= M\,d\theta \end{aligned}$$

# Work of a Force *(contd.)*




❖**Forces which do no work:**

➢reaction at a frictionless pin due to rotation of a body around the pin

➢reaction at a frictionless surface due to motion of a body along the surface

➢weight of a body with cg moving horizontally

➢friction force on a wheel moving without slipping

❖**Sum of work done by several forces may be zero**:

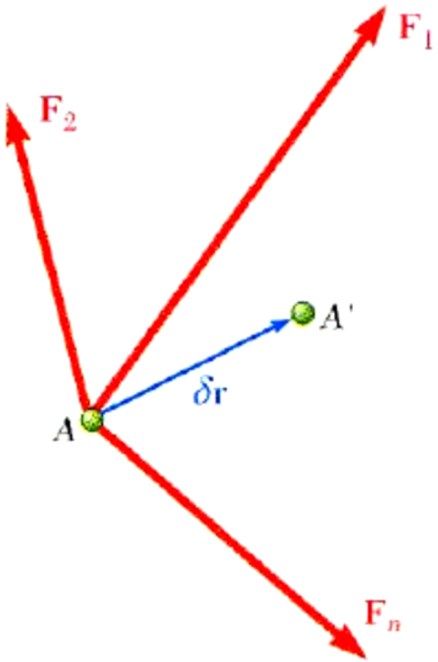
➢bodies connected by a frictionless pin

➢bodies connected by an inextensible cord

➢internal forces holding together parts of a rigid body

# Principle of Virtual Work

**Virtual Work:** Disp does not really exist but only is assumed to exist so that we may compare various possible equilibrium positions to determine the correct one.




- *Imagine* the small *virtual displacement* of particle which is acted upon by several forces.
- The corresponding *virtual work*,

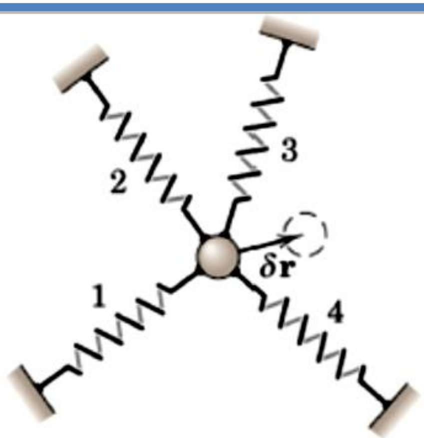
$$\delta U = \vec{F}_1 \cdot \delta\vec{r} + \vec{F}_2 \cdot \delta\vec{r} + \vec{F}_3 \cdot \delta\vec{r} = \left(\vec{F}_1 + \vec{F}_2 + \vec{F}_3\right) \cdot \delta\vec{r}$$
$$= \vec{R} \cdot \delta\vec{r}$$

*Principle of Virtual Work*:

- If a particle is in equilibrium, the total virtual work of forces acting on the particle is zero for any virtual displacement.
- If a rigid body is in equilibrium, the total virtual work of external forces acting on the body is zero for any virtual displacement of the body.
- If a system of connected rigid bodies remains connected during the virtual displacement, only the work of the external forces need be considered since work done by internal forces (equal, opposite, and collinear) cancels each other.

# Equilibrium of a Particle

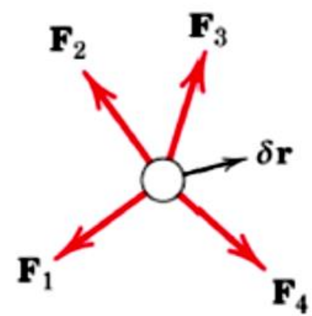
## Equilibrium of a Particle




Total virtual work done on the particle due to virtual displacement $\delta\mathbf{r}$:

$$\delta U = \mathbf{F}_1 \cdot \delta\mathbf{r} + \mathbf{F}_2 \cdot \delta\mathbf{r} + \mathbf{F}_3 \cdot \delta\mathbf{r} + \cdots = \Sigma\mathbf{F} \cdot \delta\mathbf{r}$$

Expressing $\sum\mathbf{F}$ in terms of scalar sums and $\delta\mathbf{r}$ in terms of its component virtual displacements in the coordinate directions:




$$\begin{aligned}\delta U &= \Sigma\mathbf{F} \cdot \delta\mathbf{r} = (\mathbf{i}\,\Sigma F_x + \mathbf{j}\,\Sigma F_y + \mathbf{k}\,\Sigma F_z) \cdot (\mathbf{i}\,\delta x + \mathbf{j}\,\delta y + \mathbf{k}\,\delta z) \\ &= \Sigma F_x\,\delta x + \Sigma F_y\,\delta y + \Sigma F_z\,\delta z = 0\end{aligned}$$

The sum is zero since $\sum\mathbf{F} = 0$, which gives $\sum F_x = 0$, $\sum F_y = 0$, $\sum F_z = 0$

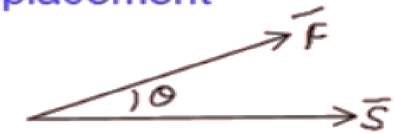
Alternative Statement of the equilibrium: $\delta U = 0$

This condition of zero virtual work for equilibrium is both necessary and sufficient since we can apply it to the three mutually perpendicular directions
→ **3 conditions of equilibrium**

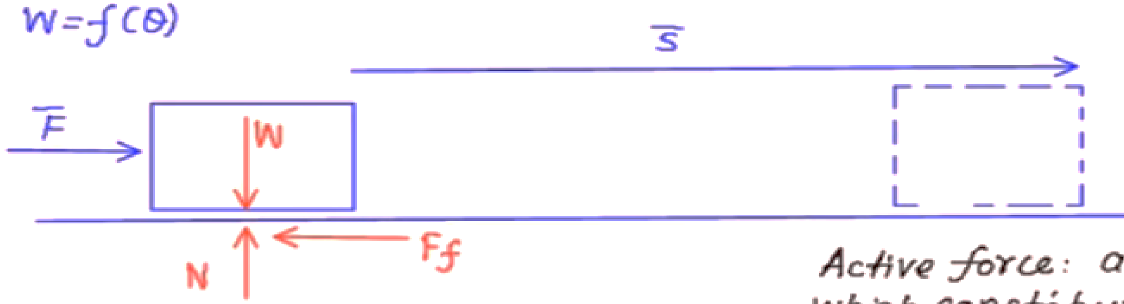
## Summery: WORK is dot product of force and displacement

$$W = \bar{F}\cdot\bar{S} = |\bar{F}|\,|\bar{S}|\cos\theta$$

$$W = f(\theta)$$




$$W = \vec{F}\vec{S}$$




Active $W_1 = \vec{F}\cdot\vec{S}\cdot\cos 0 = +FS$

Pass $\begin{cases} W_2 = W\downarrow\ \vec{S}\ \cos 90 = 0 \\ W_3 = N\uparrow\ \vec{S}\ \cos 90 = 0 \end{cases}$

Active $W_4 = \vec{F_f}\ \vec{S}\ \cos 180 = -F_f\cdot S$

Active force: are those forces which contribute in work either positive or Negative

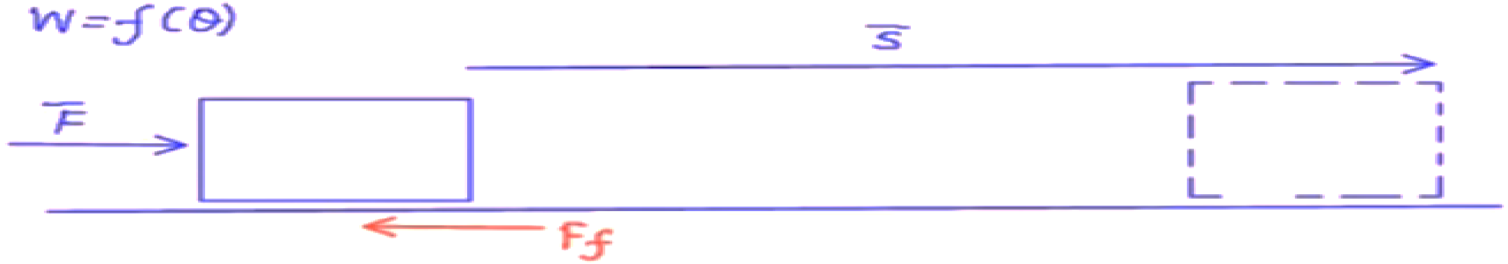
Passive Force: are those forces which do not contribute in work (Zero) are called Passive force

**Note:**

- **If the direction of force and displacement are <u>same</u> then work is <u>positive</u>**
- **If the direction of force and displacement are <u>opposite</u> then work is <u>Negative</u>**
- **If the direction of force and displacement are <u>perpendicular</u> then work is <u>zero</u>**

## Note: In Virtual work technique only active force is shown:

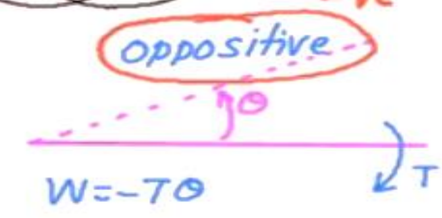
$$W = f(\theta)$$

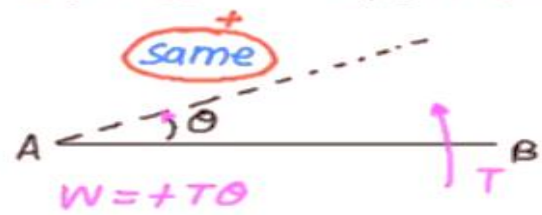


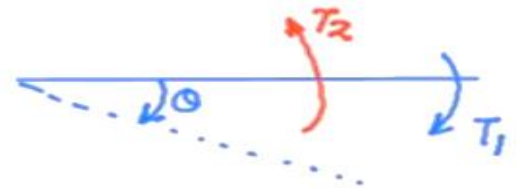
$W = \bar{F} \cdot \bar{S} = \bar{T} \cdot \bar{\theta}$

$M$ = Torque & Angular disp $= \theta$

$F$ = Force linear disp $= x, y$




$W_1 = +T_1\theta$

$W_2 = -T_2\theta$

The principle of virtual work states

*If an small imaginary displacement is given to the system under equilibrium, then sum of work done by force and moment is always equal to zero.*

*IF* $P_1, P_2, P_3, -------P_n,$ *are force and*

$\delta_1, \delta_2, \delta_3, -------\delta_n$ *are corresponding displacement*

$M_1, M_2, M_3, -------M_n$ *are moment*

*and* $\delta\theta_1, \delta\theta_2, \delta\theta_3, -------\delta\theta_n$ *are corresponding angular displacement*

*then according to principle of virtual work*

*Total virtual work done is zero*

$$P_1\delta_1 + P_2\delta_2 + -----+M_1\delta\theta_1 + M_2\delta\theta_2 + ----=0$$

$\Sigma W = 0$ —①

There are two approaches

Geometrical approach
Beams

Mathematical approach
Mechanism

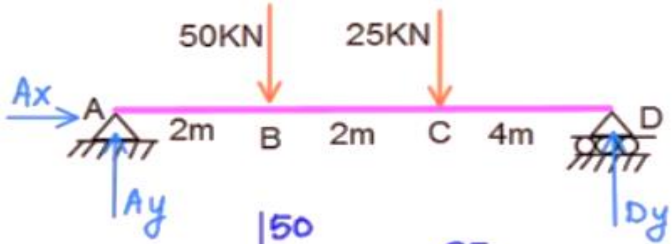
*Geometric approach: This approach is used to find reactions on beams.*

*Rules*

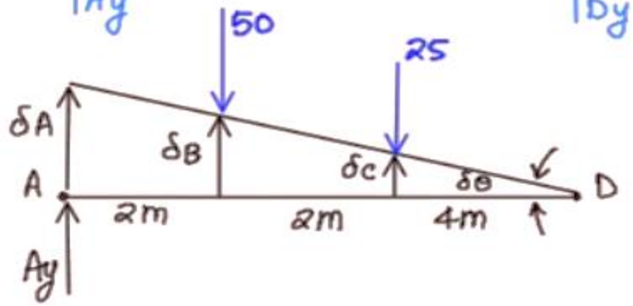
1) *The rotation of beam is permitted about the supports only ( Roller, Fixed or Hinged)*
2) *To find vertical reaction at any point, give the vertical displacement in the direction of reaction and represent only vertical forces and corresponding vertical displacement*
3) *To find horizontal reaction at any point, give the horizontal displacement in the direction of reaction and represent only horizontal forces and corresponding vertical displacement*
4) *Find unknown as principle of virtual work Sum of virtual work =0*

Procedure

1) To find the unknown reaction at the support, give the displacement in the same direction of reaction and rotate the beam about other support (i.e. other support is fixed)
2) Find unknown using Principle of virtual work




Unknown
Vertical $A_y \rightarrow D_y$
Horz $A_x$




To find $A_y$
a) show $A_y$ as assumed
b) Rotate about D

Very small angle $\tan\theta = \sin\theta = \theta$

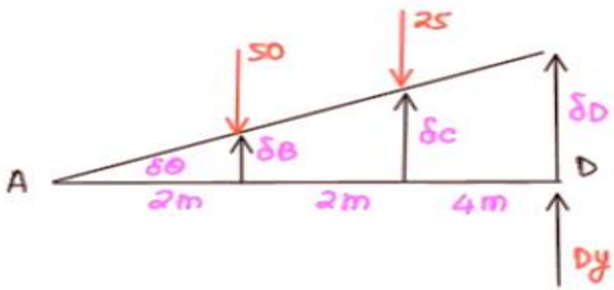
$$\tan\delta\theta = \frac{\delta c}{4} = \frac{\delta B}{6} = \frac{\delta A}{8} = \delta\theta \quad \text{—(1)}$$

$\Sigma VW = 0$

$+A_y\,\delta A - 50\,\delta B - 25\,\delta c = 0$

$A_y(8\cancel{\delta\theta}) - 50(6\cancel{\delta\theta}) - 25(4\cancel{\delta\theta}) = 0$

$A_y = 50\,kN$




To find $D_y$ Rotate about A

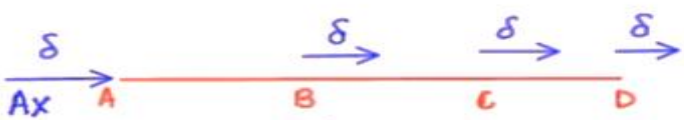
$$\tan\delta\theta = \delta\theta = \frac{\delta B}{2} = \frac{\delta c}{4} = \frac{\delta D}{8}$$

$\Sigma VW = 0$

$-50\,\delta B - 25\,\delta c + D_y \cdot \delta D = 0$

$-50(2\cancel{\delta\theta}) - 25(4\cancel{\delta\theta}) + D_y(8\cancel{\delta\theta}) = 0$

$D_y = 25\,kN$



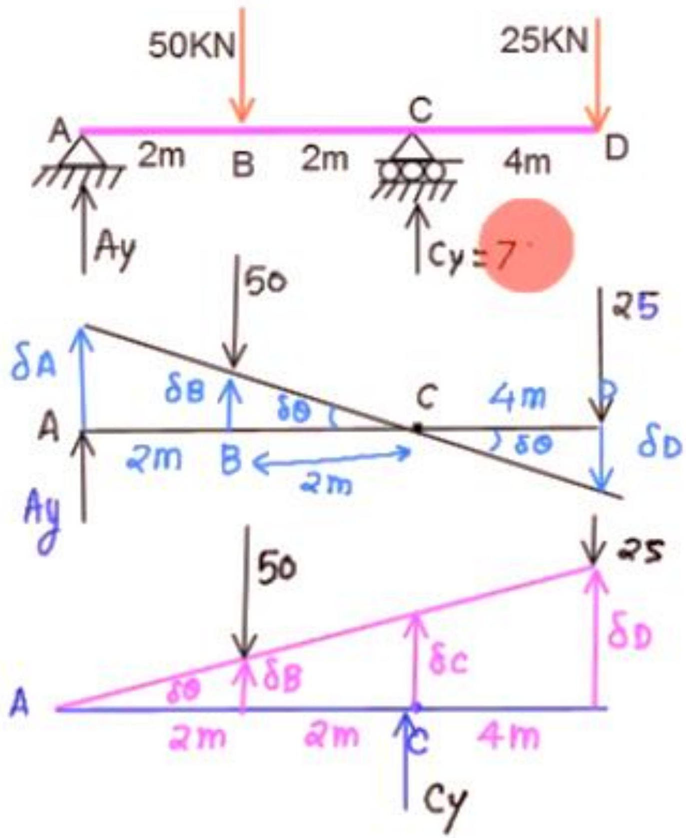

To find Horz Reaction $A_x$
Give Horz disp to Beam ABD

$\Sigma VW = 0$

$A_x \cdot \delta = 0$

$A_x = 0$

Note: As there is no horizontal force




Unknown Vertical $A_y \rightarrow C_y$

$A_y$:

$$\tan\theta = \delta\theta = \frac{\delta_D}{4} = \frac{\delta_B}{2} = \frac{\delta_A}{4}$$

$$\Sigma VW = 0$$

$$A_y\,\delta_A - 50\,\delta_B + 25\,\delta_D = 0$$

$$A_y = 0$$

$C_y$: Rotate about A

$$\tan\theta = \delta\theta = \frac{\delta_B}{2} = \frac{\delta_c}{4} = \frac{\delta_D}{8}$$

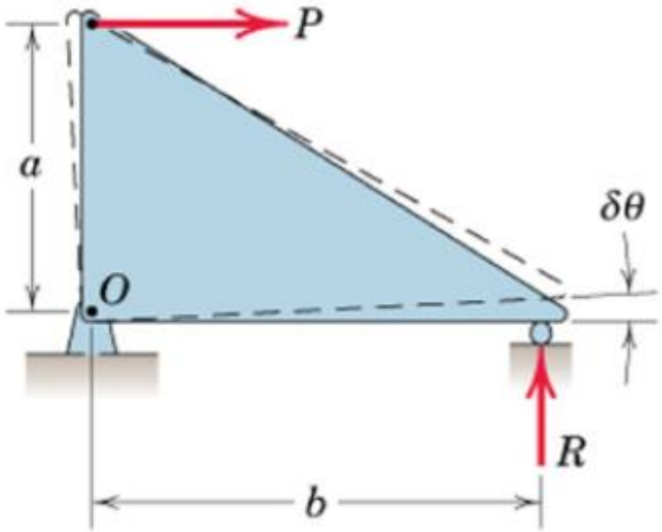
$$\Sigma VW = 0$$

$$-50\,\delta_B + C_y\,\delta_c - 25\,\delta_D = 0$$

$$-50 \times 2\delta\theta + C_y(4\delta\theta) - 25(8\delta\theta) = 0$$

$$C_y = 75$$

# Applications of Principle of Virtual Work




## Equilibrium of a Rigid Body

Total virtual work done on the entire rigid body is zero since virtual work done on each Particle of the body in equilibrium is zero.

Weight of the body is negligible.

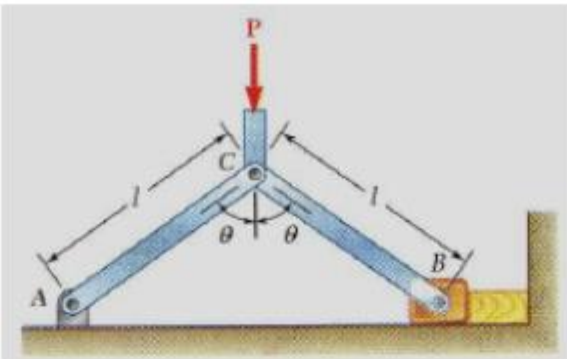
Work done by $P = -Pa\,\delta\theta$
Work done by $R = +Rb\,\delta\theta$

Principle of Virtual Work: $\delta U = 0$:
$-Pa\,\delta\theta + Rb\,\delta\theta = 0$

→ $Pa - Rb = 0$
→ Equation of Moment equilibrium @ O.
→ Nothing gained by using the Principle of Virtual Work for a single rigid body

# Problem 1

Determine the force exerted by the vice on the block when a given force $P$ is applied at C. Assume that there is no friction.



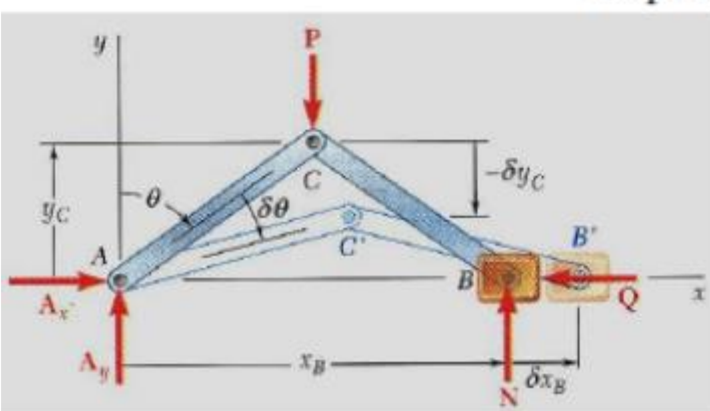

- Consider the work done by the external forces for a virtual displacement $\delta\theta$. $\delta\theta$ is a positive increment to $\theta$ in bottom figure. Only the forces $P$ and $Q$ produce nonzero work.
- $x_B$ increases while $y_C$ decreases

→ +ve increment for $x_B$: $\delta x_B$ → $\delta U_Q = -Q\delta x_B$ (opp. Sense)

→ -ve increment for $y_C$: $-\delta y_C$ → $\delta U_P = +P(-\delta y_C)$ (same Sense)

$$\delta U = 0 = \delta U_Q + \delta U_P = -Q\,\delta x_B - P\,\delta y_C$$

Expressing $x_B$ and $y_C$ in terms of $\theta$ and differentiating w.r.t. $\theta$




$$x_B = 2l\sin\theta \qquad y_C = l\cos\theta$$

$$\delta x_B = 2l\cos\theta\,\delta\theta \qquad \delta y_C = -l\sin\theta\,\delta\theta$$

$$0 = -2Ql\cos\theta\,\delta\theta + Pl\sin\theta\,\delta\theta$$

$$Q = \tfrac{1}{2}P\tan\theta$$

By using the method of virtual work, all unknown reactions were eliminated. $\sum M_A$ would eliminate only two reactions.

- If the virtual displacement is consistent with the constraints imposed by supports and connections, only the work of loads, applied forces, and friction forces need be considered.

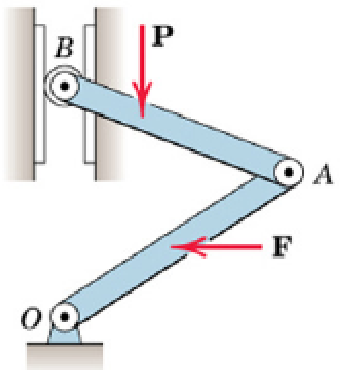
❖ **Note:** If the force and displacement direction are opposite then (-)ve work done.

# Principle of Virtual Work *(contd.)*

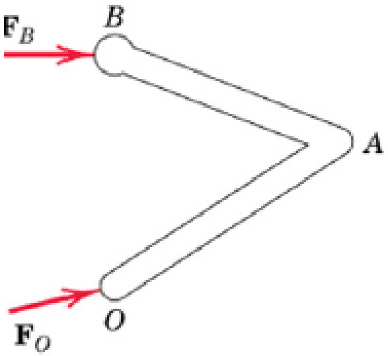
Virtual Work done by external active forces on an ideal mechanical system in equilibrium is zero for any and all virtual displacements consistent with the constraints

$$\delta U = 0$$

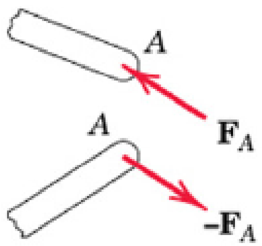
Three types of forces act on interconnected systems made of rigid members




Active Forces: Work Done
Active Force Diagram




Reactive Forces
No Work Done




Internal Forces
No Work Done

# Major Advantages of Virtual Work Method

- It is not necessary to dismember the systems in order to establish relations between the active forces.
- Relations between active forces can be determined directly without reference to the reactive forces.

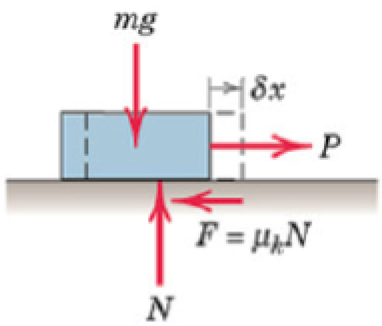
→ The method is particularly useful in determining the position of equilibrium of a system under known loads (This is in contrast to determining the forces acting on a body whose equilibrium position is known – studied earlier).

→ The method requires that internal frictional forces do negligible work during any virtual displacement.

→ If internal friction is appreciable, work done by internal frictional forces must be included in the analysis.

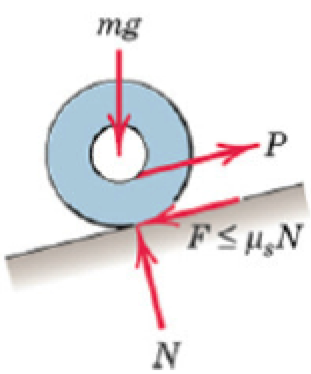
# Principle of Virtual Work application to systems having Friction

- So far, the Principle of virtual work was discussed for "ideal" systems.
- If significant friction is present in the system ("Real" systems), work done by the external active forces (input work) will be opposed by the work done by the friction forces.




During a virtual displacement $\delta x$:

Work done by the kinetic friction force is: $-\mu_k N \delta x$



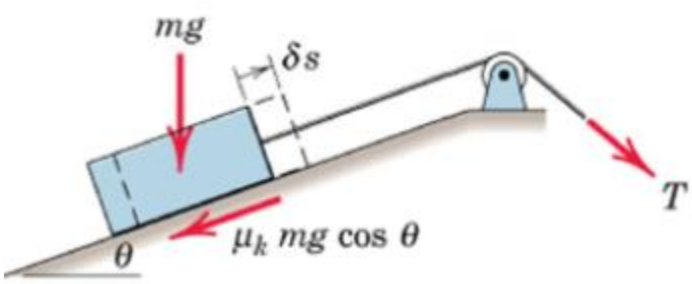

During rolling of a wheel:
the static friction force does no work if the wheel does not slip as it rolls.

# Mechanical Efficiency(e)

- Output work of a machine is always less than the input work because of energy loss due to friction.

$$e = \frac{\text{Output Work}}{\text{Input Work}}$$

For simple machines with SDOF & which operates in uniform manner, mechanical efficiency may be determined using the method of Virtual Work




For the virtual displacement $\delta s$: Output Work is that necessary to elevate the block = $mg\ \delta s\ sin\theta$

Input Work = $T\ \delta s = mg\ sin\theta\ \delta s + \mu_k\ mg\ cos\theta\ \delta s$

The efficiency of the inclined plane is:

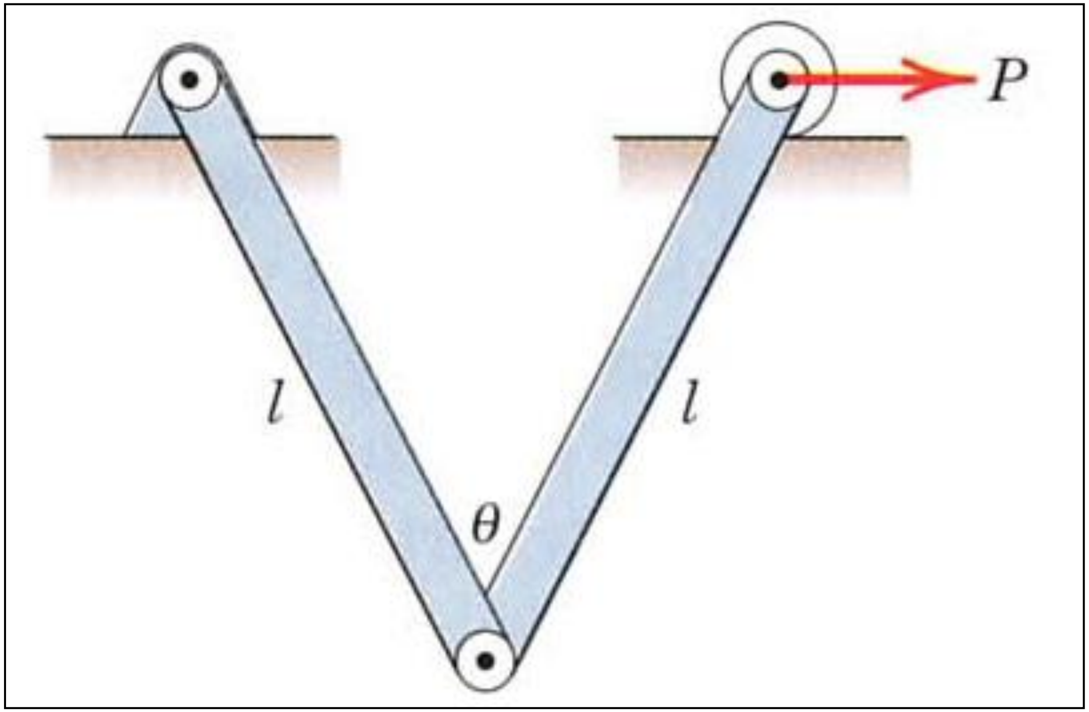
$$e = \frac{mg\delta s \sin\theta}{mg(\sin\theta + \mu_k \cos\theta)\delta s} = \frac{1}{1+\mu_k \cot\theta}$$

As friction decreases, Efficiency approaches unity the efficiency of the jack can be expressed as

$$e = \frac{Wr\ \delta\theta \tan \alpha}{Wr\ \delta\theta \tan (\alpha + \phi)} = \frac{\tan \alpha}{\tan (\alpha + \phi)}$$

# Problem 2

Each of the two uniform hinged bars has a mass $m$ and a length $l$, and is supported and loaded as shown. For a given force $P$ determine the angle $\theta$ for equilibrium.



# Problem 2 *(contd.)*

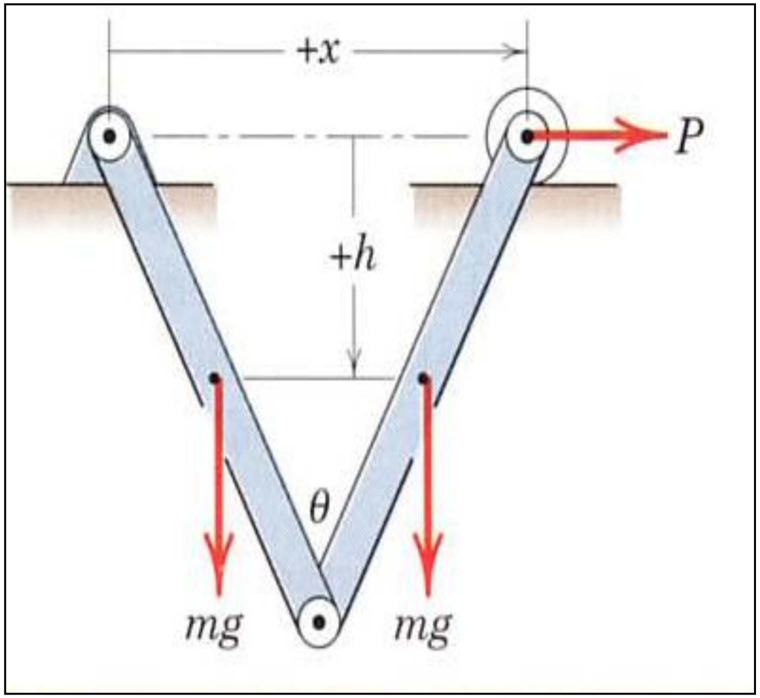
**Solution.** The active-force diagram for the system composed of the two members is shown separately and includes the weight $mg$ of each bar in addition to the force $P$. All other forces acting externally on the system are reactive forces which do no work during a virtual movement $\delta x$ and are therefore not shown.

$$P\,\delta x + 2mg\,\delta h = 0$$

$$x = 2l\sin\frac{\theta}{2} \quad \text{and} \quad \delta x = l\cos\frac{\theta}{2}\,\delta\theta$$

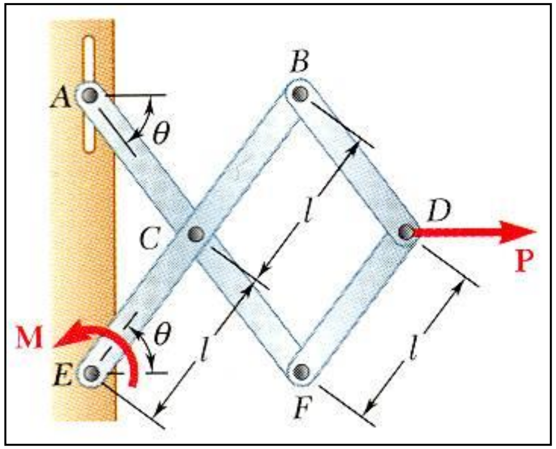
$$h = \frac{l}{2}\cos\frac{\theta}{2} \quad \text{and} \quad \delta h = -\frac{l}{4}\sin\frac{\theta}{2}\,\delta\theta$$

$$Pl\cos\frac{\theta}{2}\,\delta\theta - 2mg\frac{l}{4}\sin\frac{\theta}{2}\,\delta\theta = 0$$

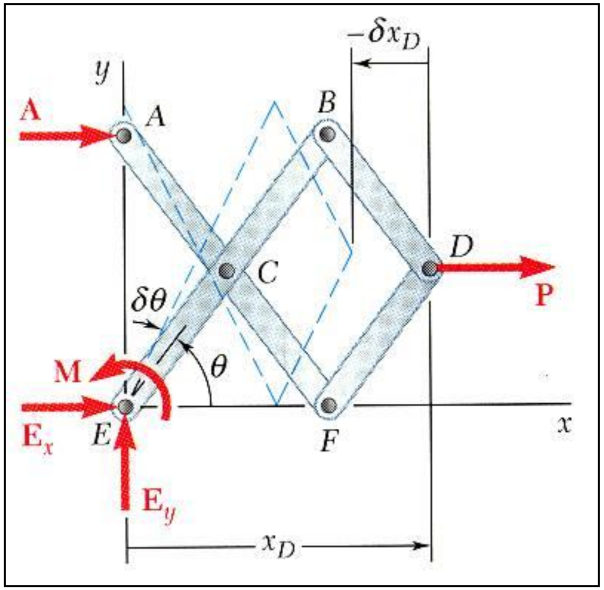



$$\tan\frac{\theta}{2} = \frac{2P}{mg} \quad \text{or} \quad \theta = 2\tan^{-1}\frac{2P}{mg}$$

# Problem 3




**Fig(a)**




- Determine the magnitude of the couple $M$ required to maintain the equilibrium of the mechanism shown in **Fig(a)**?

**SOLUTION**:

- Apply the principle of virtual work,

$$\delta U = 0 = \delta U_M + \delta U_P$$
$$0 = M\delta\theta + P\delta x_D$$

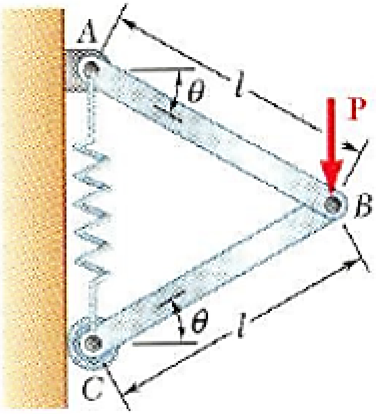
$$x_D = 3l\cos\theta$$
$$\delta x_D = -3l\sin\theta\delta\theta$$

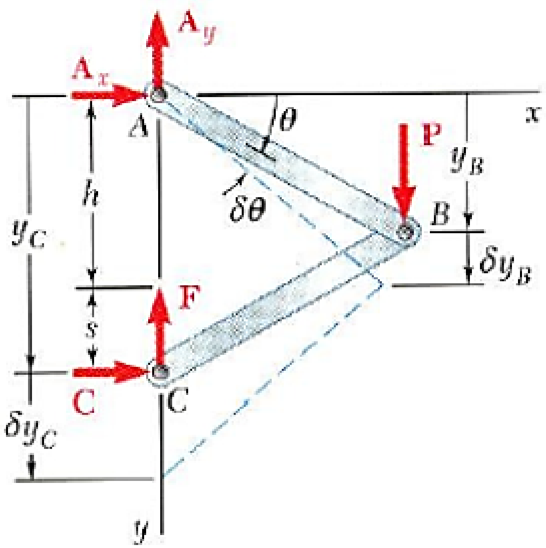
$$0 = M\delta\theta + P(-3l\sin\theta\delta\theta)$$

$$\boxed{M = 3Pl\sin\theta}$$

# Problem 4

## Example




Determine the expressions for $\theta$ and the tension in the spring which correspond to the equilibrium position of the spring. The unstretched length of the spring is *h* and the constant of the spring is k. Neglect the weight of the mechanism.

SOLUTION:

- Apply the principle of virtual work

$$\delta U = \delta U_B + \delta U_F = 0$$

$$0 = P\delta y_B - F\delta y_C$$

$$y_B = l\sin\theta \qquad y_C = 2l\sin\theta \qquad F = ks$$

$$\delta y_B = l\cos\theta\delta\theta \qquad \delta y_C = 2l\cos\theta\delta\theta \qquad = k(y_C - h)$$

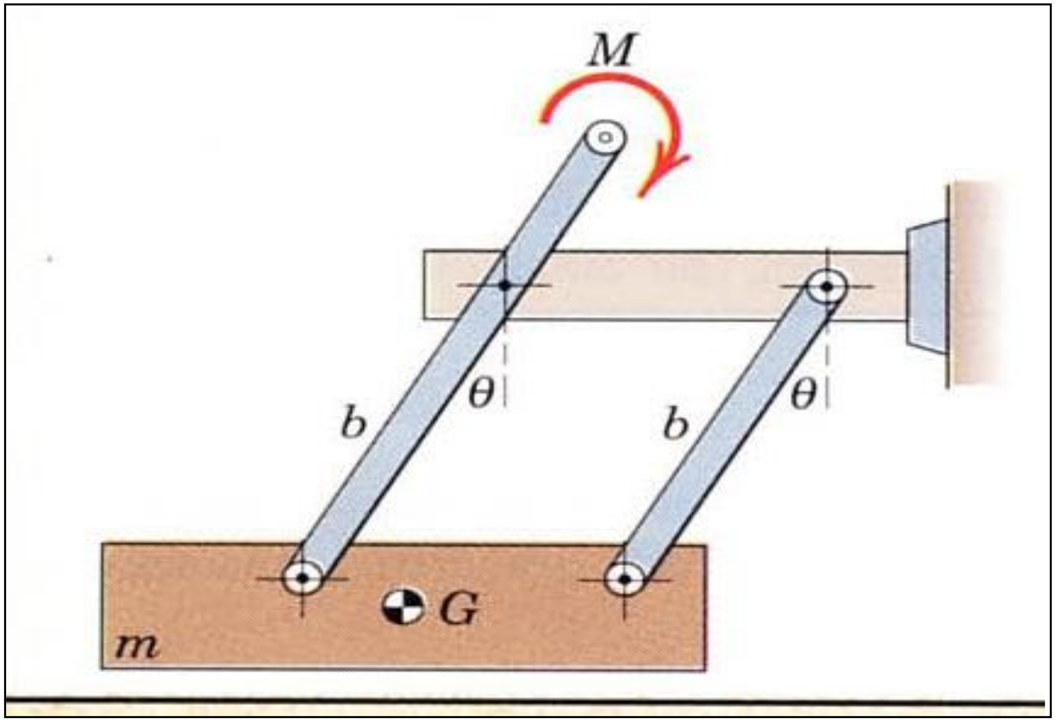
$$= k(2l\sin\theta - h)$$

$$0 = P(l\cos\theta\delta\theta) - k(2l\sin\theta - h)(2l\cos\theta\delta\theta)$$

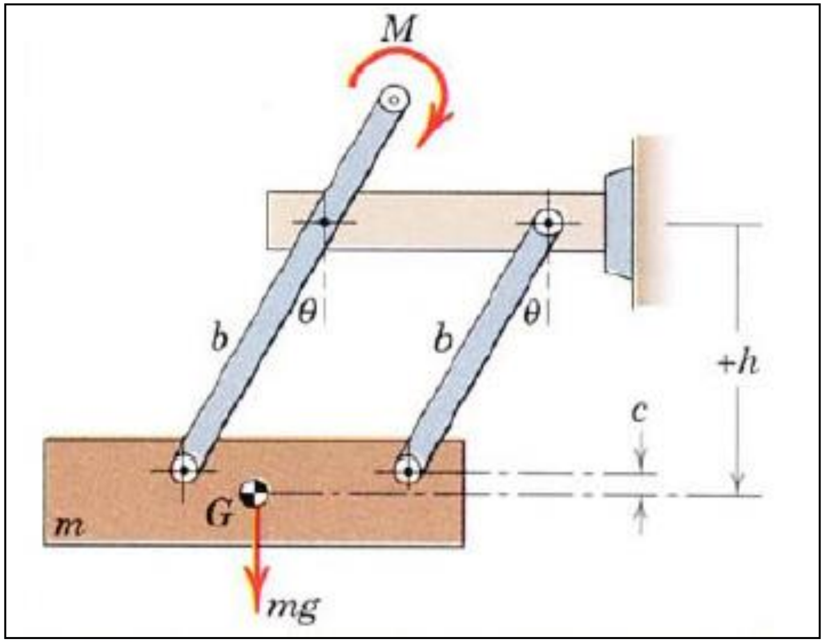
$$\boxed{\begin{array}{c} \sin\theta = \dfrac{P + 2kh}{4kl} \\ F = \tfrac{1}{2}P \end{array}}$$

# Problem 5

The mass $m$ is brought to an equilibrium position by the application of the couple $M$ to the end of one of the two parallel links which are hinged as shown. The links have negligible mass, and all friction is assumed to be absent. Determine the expression for the equilibrium angle $\theta$ assumed by the links with the vertical for a given value of $M$. Consider the alternative of a solution by force and moment equilibrium.



# Problem 5 *(contd.)*




$$h = b\cos\theta + c.$$

$$+mg\,\delta h = mg\,\delta(b\cos\theta + c)$$
$$= mg(-b\sin\theta\,\delta\theta + 0)$$
$$= -mgb\sin\theta\,\delta\theta$$

$[\delta U = 0]$ $$M\,\delta\theta + mg\,\delta h = 0$$

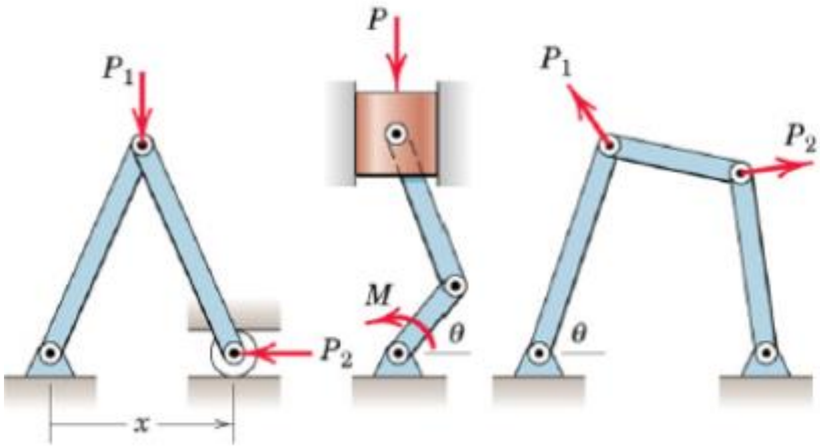
which yields

$$M\,\delta\theta = mgb\sin\theta\,\delta\theta$$

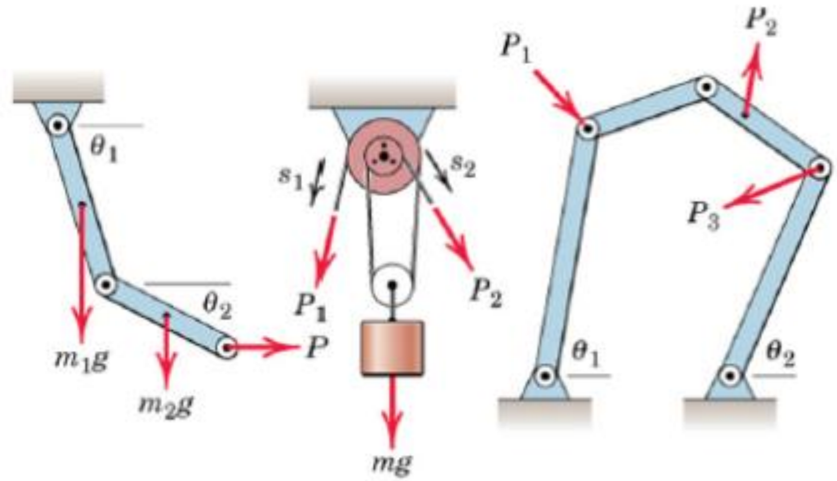
$$\theta = \sin^{-1}\frac{M}{mgb}$$

# Degrees of Freedom *(DOF)*

- Number of independent coordinates needed to specify completely the configuration of system




(a) Examples of one-degree-of-freedom systems




(b) Examples of two-degree-of-freedom systems

Only one coordinate (displacement or rotation) is needed to establish position of every part of the system

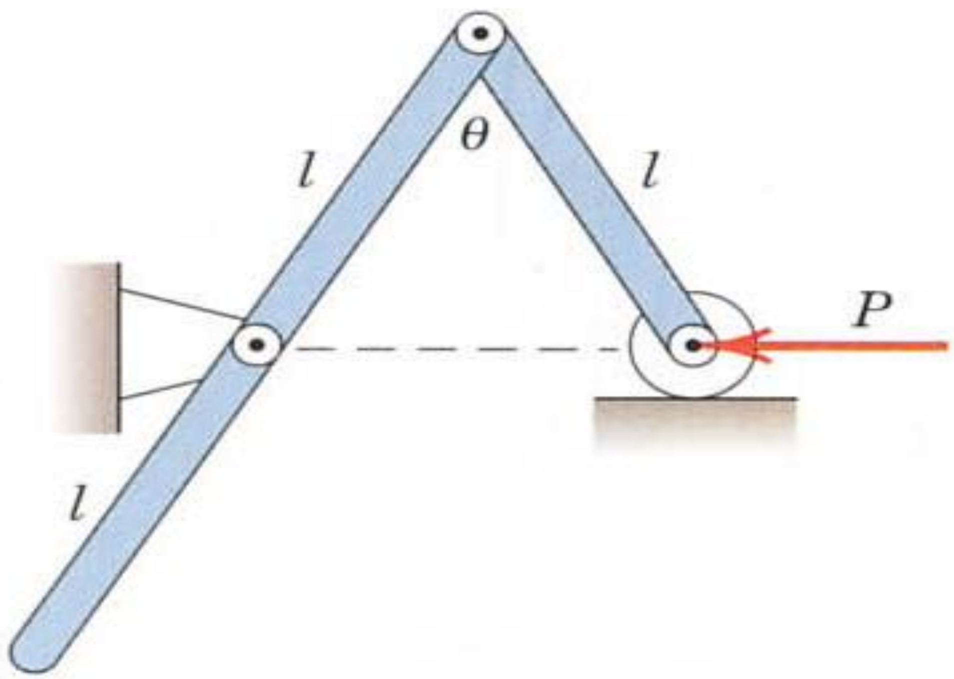
Two independent coordinates are needed to establish position of every part of the system

$\delta U=0$ can be applied to each DOF at a time keeping other DOF constant.
ME101 → only SDOF systems

# END

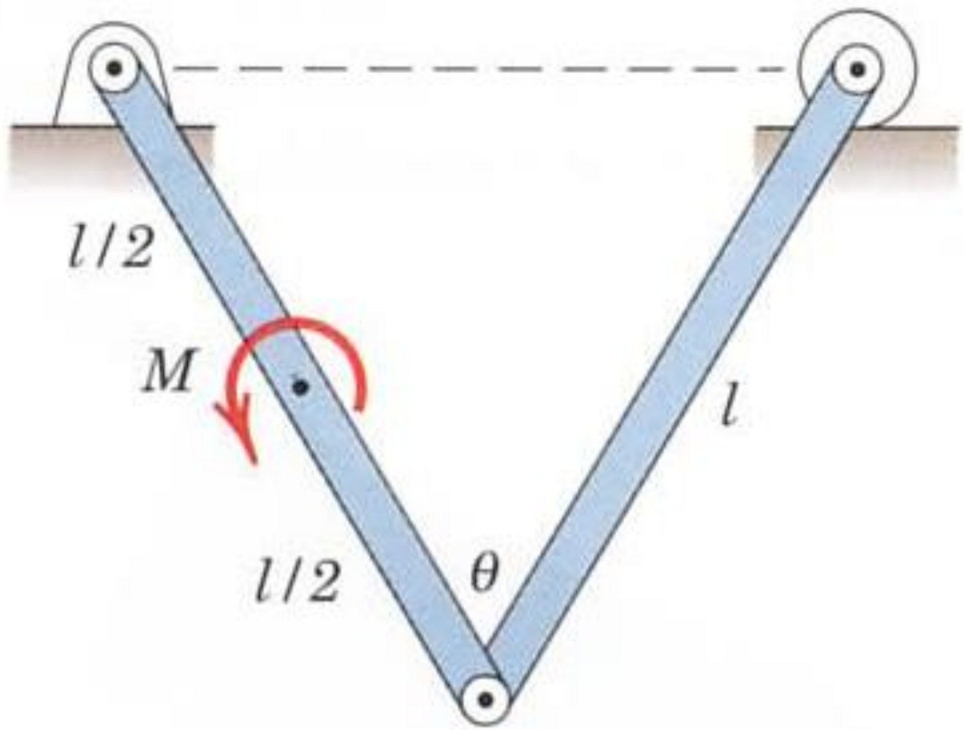
# Assignments

**7/1** The mass of the uniform bar of length $l$ is $m$ while that of the uniform bar of length $2l$ is $2m$. For a given force $P$, determine the angle $\theta$ for equilibrium.




**Problem 7/1**

**7/2** Determine the couple $M$ required to maintain equilibrium at an angle $\theta$. Each of the two uniform bars has mass $m$ and length $l$.




**Problem 7/2**

**/4** The spring of constant $k$ is unstretched when $\theta = 0$. Derive an expression for the force $P$ required to deflect the system to an angle $\theta$. The mass of the bars is negligible.